

ECE 315

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Lecture 31 :

- Motion control
- Encoders
- Servo and DC Motors
- Other actuators



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Last Lecture

- Motion control
- Stepper motors
- HW controllers, H-bridge

This Lecture

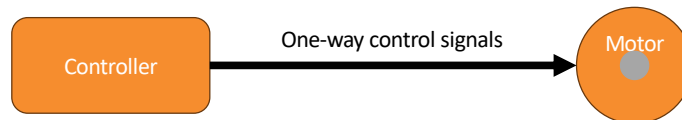
- Motion control
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- Servo and DC Motors
- Other actuators

Learning Objectives

- Explain and analyze open-loop and closed-loop motion control systems using encoders and feedback
- Describe and apply principles of operation for servo motors, brushed and brushless DC motors, including PWM and H-bridge control
- Select and justify appropriate electro-mechanical actuators (motors, linear actuators, solenoids, relays) for embedded system applications
- Distinguish between open-loop and closed-loop motion control systems and explain their advantages, limitations, and typical use cases
- Describe the principles of operation of rotary shaft encoders, including optical, magnetic, and magneto-resistive types, and explain how encoder feedback enables closed-loop control
- Identify and compare other electro-mechanical actuators, including linear actuators, solenoids, and mechanical versus solid-state relays, with respect to functionality and application domains

General Motor Motion Control

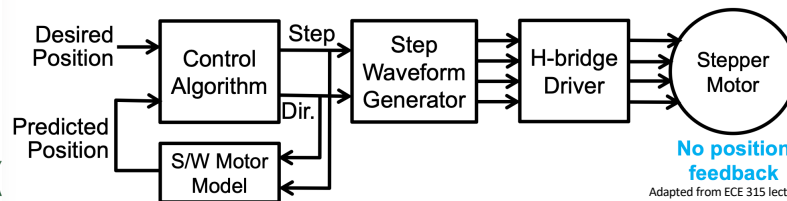
- In the previous lecture we explored how a stepper motor is constructed and works, and how to drive it
- We never specified how we might know the current position of the stator (assume here that the stator is attached to the motor shaft)
- This form of control is known as “open loop”
 - There is no knowledge of the motor state (position, speed, power, ...) that is returned (aka, “fed back”) to the controller



Open-loop Control

- In open-loop control, there is no knowledge of the motor state (position, speed, power, ...) that is returned (aka, “fed back”) to the controller
- Sometimes it doesn't really matter...
- If needed, it is estimated using some sort of model, e.g., manually step the motor to a starting position, then count the steps made and in which direction
 - This might be “good enough”

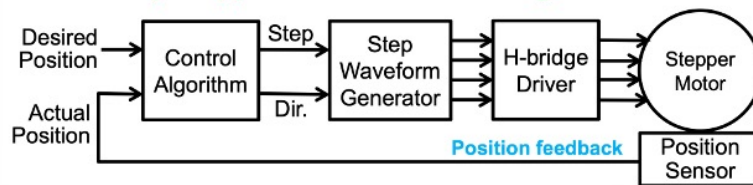
Open-loop Stepper Motor Control Configuration:



Closed-loop Control

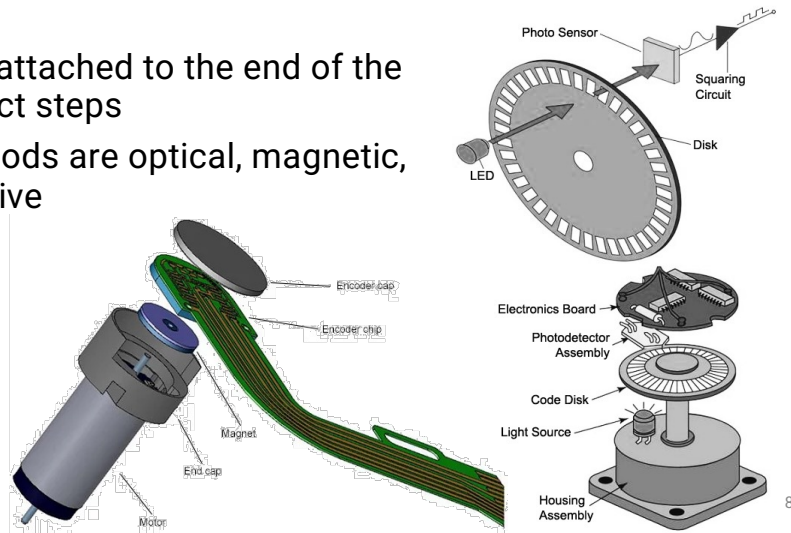
- In closed-loop control, the motor state (position, speed, power, ...) is directly measured and returned to the controller to aid in motor control
- The control algorithm is generally digital, which means that analogue signals from the position or other sensor must be converted.
- The control algorithm computes the difference between the desired position and the actual position, then determines the best next step command(s)

Closed-loop Stepper Motor Control Configuration:



Rotary Shaft Encoders

- An encoder can be attached to the end of the motor shaft to detect steps
- Three popular methods are optical, magnetic, and magneto-resistive

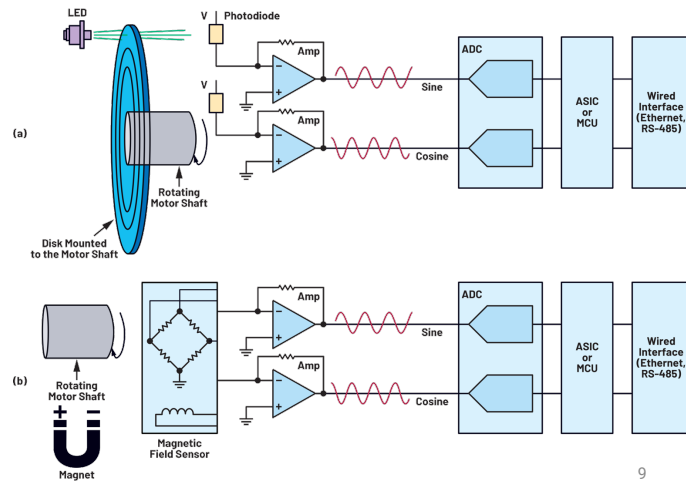


<https://www.portescap.com/en/products/accessories/workings-of-encoders>

https://anaheimautomation.com/blog/post/encoders-guide?srsltid=AfmBOopSN8QsEVUB5npwG74pPuTYLMUKpIXT1z_QgFCU_BcXHBSQkxUv

Rotary Shaft Encoders

- Optical encoders rely on a wheel with slots and counting when detected light changes
- Either pulses or varying intensities indicate the position using sine and cosine relations
 - 90° phase shifted varying intensity is shown in (a)
- A magneto-resistive uses a bipolar magnet and sine-cosine measurements to determine the angle of rotation

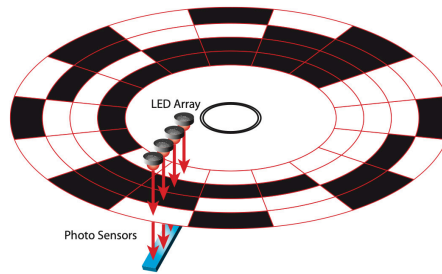


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<https://www.analog.com/en/resources/analog-dialogue/articles/how-to-design-motor-encoders-for-next-generation-sustainable-applications.html>

Gray Code Optical Encoder

- An array of light sources can be used to uniquely detect shaft rotations using a binary code
- A “Gray” code is one designed so that for each step only one bit changes
- This makes it easy to detect errors and simplifies the digital logic

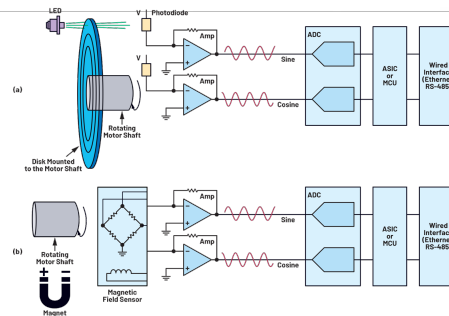
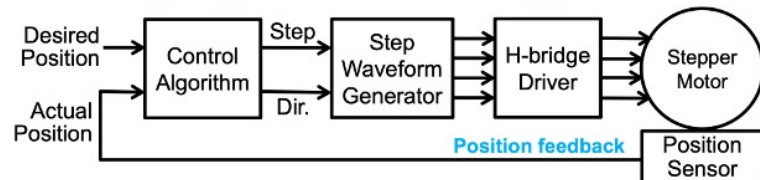


Position	Binary	Gray
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

Close-loop Control, again

- Encoders provide position information that can be used in the closed-loop, feedback control system

Closed-loop Stepper Motor Control Configuration:



Servos, DC Motors

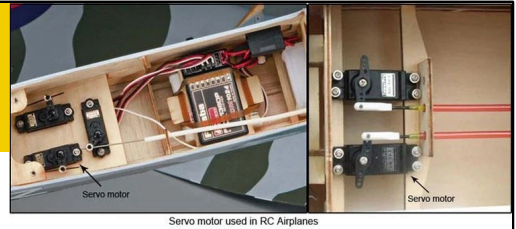


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Servos

- A **servo motor** is an electromechanical actuator designed for precise control of position, velocity, or torque using closed-loop feedback.
 - Used in radio-controlled aircraft to control the flight surfaces: rudder (yaw control), elevator (pitch control), and ailerons (roll control).
- **Closed-loop control:** A feedback device (typically a potentiometer or encoder) continuously measures the output and compares it to a commanded setpoint.
- **Integrated or Component system:** Usually consists of a motor (DC, AC, or brushless), a gear train, a feedback sensor, and a control circuit.
- **Commanded motion:** Moves to and holds a specific position or follows a trajectory, correcting errors automatically.
- High accuracy & repeatability: Well-suited for applications requiring precise motion



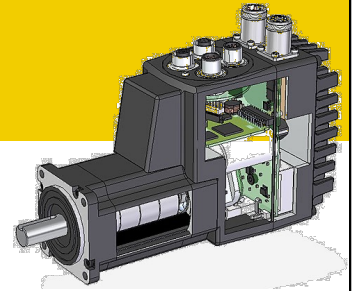
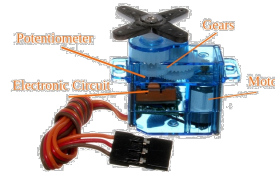
Servos cont'd

- A widely used control interface for servos uses an analog signaling method called **pulse width modulation** (PWM)
 - Many microcontrollers have built-in PWM peripherals
 - Zynq-7000 has 6 timers that can generate PWM signals at MIO pins
 - RP2040 has up to 16 controllable PWM outputs
- A train of positive pulses at a fixed rate, e.g., at 50 Hz
- Pulses are fixed amplitude (e.g., 3 to 5 V)
- Width is controlled to be within some range about a neutral width (e.g., 1.5 ms $\pm$ 0.5 ms).
- The pulse width determines the angle of a rotor

Servos cont'd

Common forms

- RC/industrial servos
 - Integrated motor and controller
 - Accept a control signal (e.g., PWM) that specifies target position
- Servo drives + motors
 - Separate motor and controller
 - Used in CNC machines, robotics, and automation



Adapted from ECE 315 lecture notes by Bruce Cockburn 15

<https://www.designworldonline.com/400w-mac400-integrated-ac-servo-motor/>

https://www.solomotorcontrollers.com/blog/servo-motor/?srsltid=AfmBOooZsp4cU6FtqPx_hbmeTVFnIdS-j7b0AhTTts4779YWQ9n6rcIX

https://anaheimautomation.com/blog/post/servo-motor-guide?srsltid=AfmBOop8jeNTpp2k_V2zx0vN8m6qJqzKkxntFCpX3rDRiK5yMu2xm4mQ

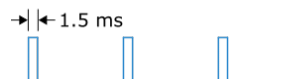
<https://www.candrian.gr/index.php/reverse-engineering-sewing-servo-motor-controller/>

Example – PWM Signal for Rotor Control

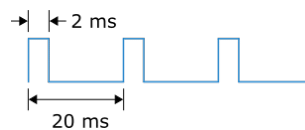
- Min and max pulse width can vary for different manufacturers
 - Min can be 0.5 to 0.8 ms
 - Max can be 2.5 to 3 ms
- Frequency usually 50 Hz
 - Can be 30 to 200 Hz
- Voltage level can vary from 2.5V to 10V



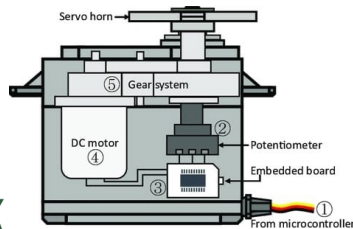
0 degrees



90 degrees



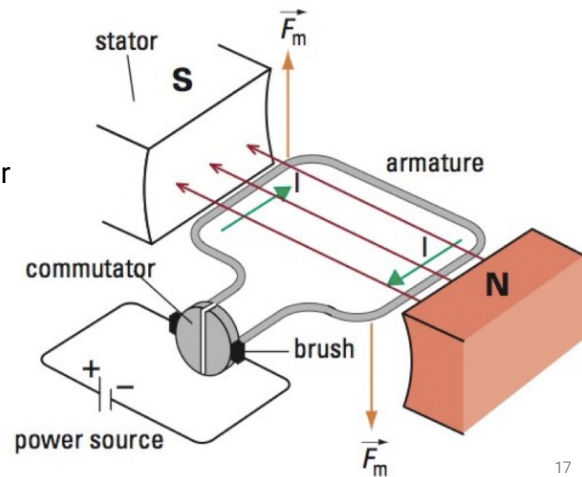
180 degrees



<https://fabacademy.org/2022/labs/bhutan/students/anith-ghalley/assignments/week13/> 16

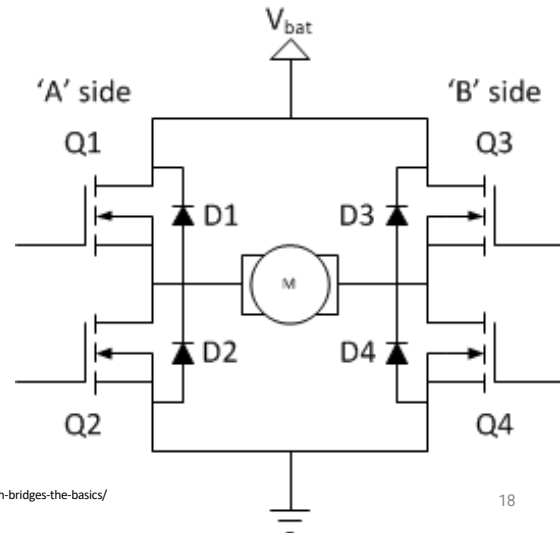
Brushed DC Motor

- DC motors where current is provided to the rotor coil via contacts on a commutator are known as “brushed”
- The motor is constructed so that when current flows a field opposing the stator (permanent) magnetic field is induced which creates a torque that turns the rotor
- The split commutator ensures the current is reversed every 180° so that the rotation is continued



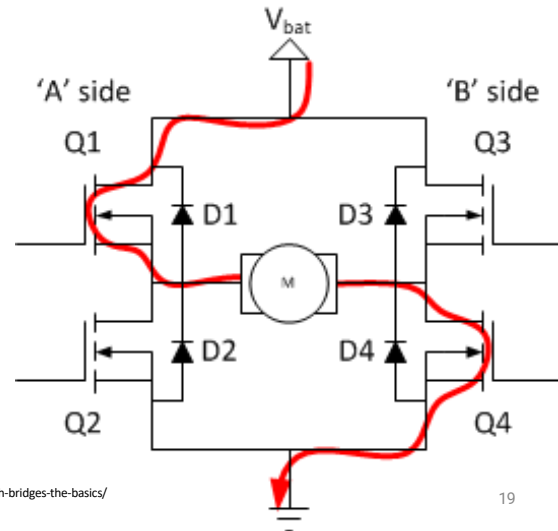
H-Bridge Control of a DC Motor

- Similar to stepper motor control, an H-bridge circuit can be used to energize the coils of a DC motor



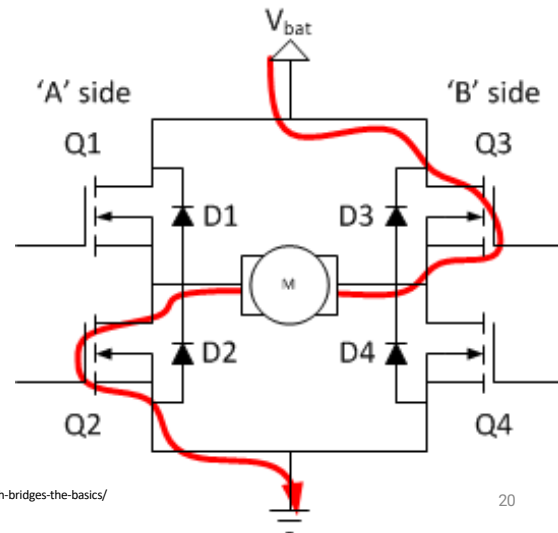
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- With **Q1** and **Q4** turned on, **Q2** and **Q3** turned off, the motor will turn in one direction



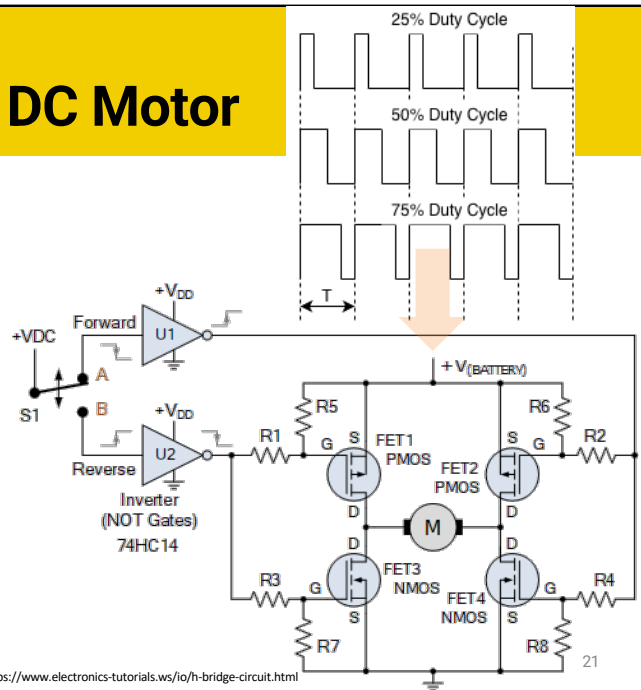
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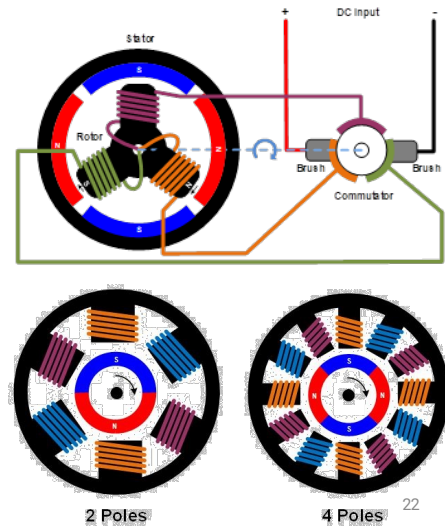
H-Bridge Control of a DC Motor

- This circuit illustrates the forward and reverse control via switch S1
- Motor speed is accomplished via pulse-width modulation (PWM)
- As with stepper motors, an encoder will help with closed-loop control of the motor
- Unlike the stepper, the motor position cannot be simply held static



Brushless DC Motor

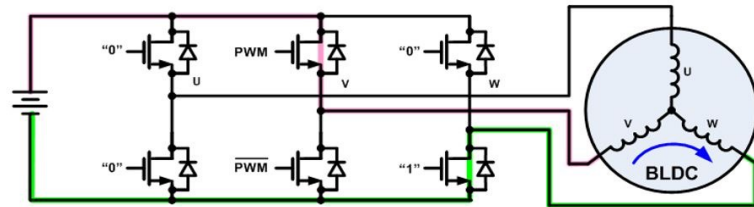
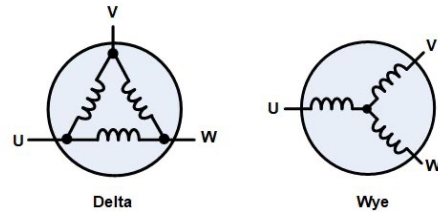
- A brushless DC motor eliminates the need for a brushed commutator by inverting the motor configuration
- Instead of the rotor having two or more coiled arms, the stator has the coils
- The rotor is comprised of two or more permanent magnets



<https://www.monolithicpower.com/en/learning/resources/brushless-vs-brushed-dc-motors?srsId=AfmBOoqaqVwTXV10uVWPA0wujiOcmhWy3lQr-jryH9shzZirxt1f7Hux>

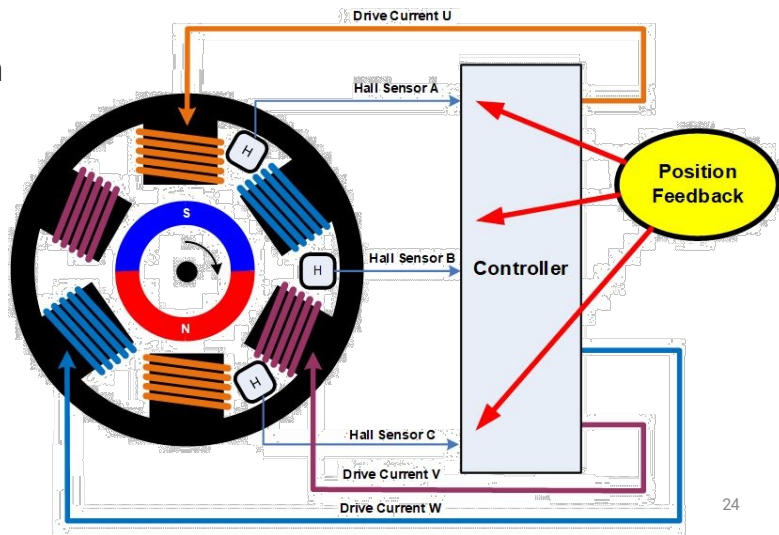
Brushless DC Motor

- The stator coils are typically connected in a Delta or Wye configuration
- The driving circuit is like an H-Bridge, but has switches (transistors) through which a PWM signal is applied to control the motor speed



Brushless DC Motor

- Hall-effect sensors can be embedded in the motor housing to provide position information
- This is used in a closed-loop control system



Other Actuators



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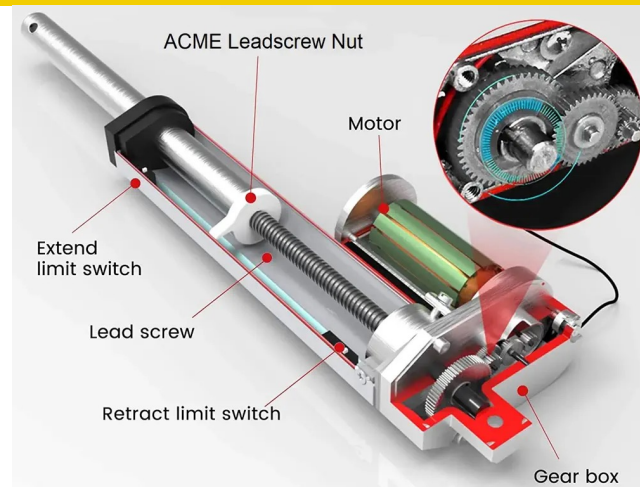
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Other Actuators

- In this course we restrict ourselves to electro-mechanical actuators
 - We don't consider hydraulic or pneumatic, for example
- Linear Actuators
- Mechanical or solid-state switches/relays
- Solenoids

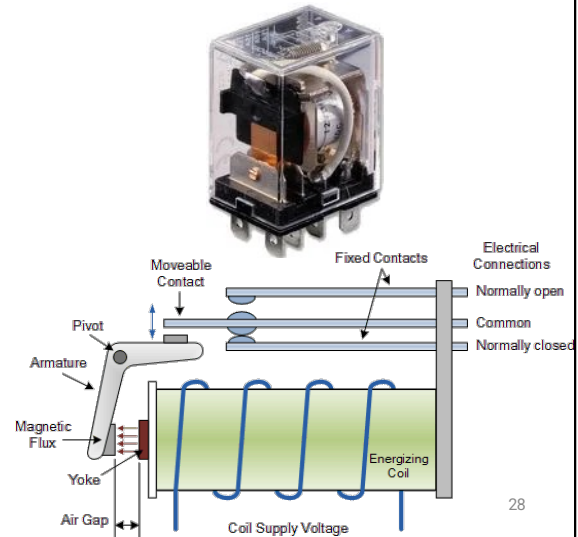
Linear Actuator

- Converts rotary motion into straight-line, linear motion
 - For rotary motion, we consider only electric motors, but gas or diesel can be used, for example.
- Used to push, pull, lift, or position loads with controlled force, speed, and displacement in electromechanical systems



Mechanical Switches/Relays

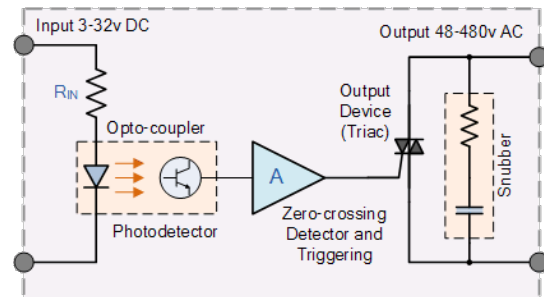
- A mechanical relay uses a magnetic coil to change the state of a mechanical electrical switch
- Typically, the relay has a configuration of contacts to a common conductor that are either normally-open or normally-closed
- This provides flexibility for the logic of the switch
- The coil circuit is physically (electrically) isolated from the switching circuit
 - Means that the switching circuit can be high-voltage and/or high-current



https://www.electronics-tutorials.ws/io/io_5.html

Solid-state Switches/Relays

- Relay is constructed using semiconductor-based devices
- Like the mechanical relay, the control circuit is electrically isolated from the switched circuit
- Solid-State Relay (SSR)
- In terms of mechanical stresses like vibration, shock, and environmental effects, SSRs are more robust than mechanical relays



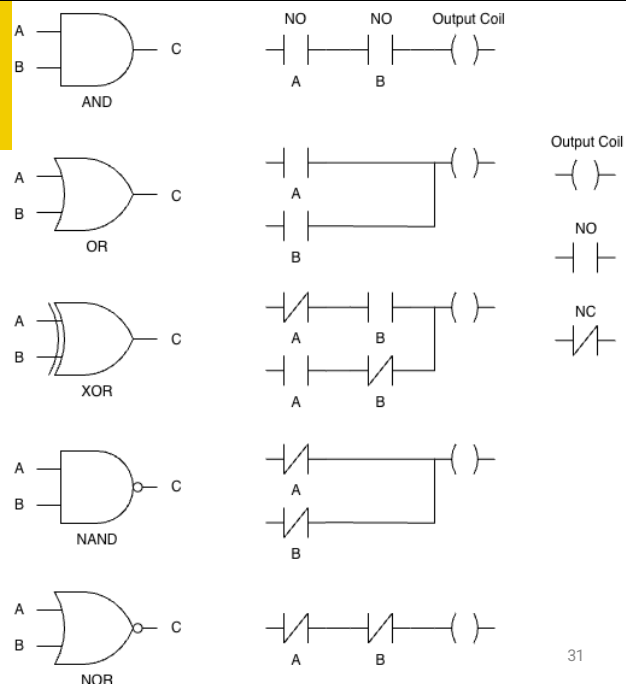
Relay/Ladder Logic Symbols

- We do not get to look at relay/ladder logic in this course
- It is a whole sub-discipline and most commonly associated with process control in plants and buildings

CIRCUIT BREAKER		DISCONNECT SWITCH		OVERLOAD HEATER ELEMENTS		AC MOTORS		DC MOTOR	
3 POLE	1 POLE	3 POLE	1 POLE	3 POLE	1 POLE	3 PHASE	1 PHASE	ARMATURE	FIELD
RETROREFLECTIVE PHOTOCELL		PROXIMITY SWITCH		LIMIT SWITCH		FLOAT SWITCH			
FUSE	POWER SOURCE	N.O. CONTACT	N.C. CONTACT	N.O. CONTACT	N.C. CONTACT	N.O. CONTACT	N.C. CONTACT	N.O. CONTACT	N.C. CONTACT
VACUUM & PRESSURE SWITCH		TEMPERATURE SWITCH		FLOW SWITCH (WATER, AIR, ETC.)		FOOT SWITCH		TOGGLE SWITCH	
N.O. CONTACT	N.C. CONTACT	N.O. CONTACT	N.C. CONTACT	N.O. CONTACT	N.C. CONTACT	N.O. CONTACT	N.C. CONTACT	N.O. CONTACT	N.C. CONTACT
SELECTOR SWITCH		PUSHBUTTONS		CONTACTS		COIL			
2 POSITION (2 CONTACT)	3 POSITION (3 CONTACT)	N.O. CONTACT	N.C. CONTACT	2 CONTACT (N.O.)	MUSHROOM HEAD N.C. CONTACT	MUSHROOM HEAD N.C. CONTACT	RELAY, CONTACTOR, STARTER	N.C. CONTACT	N.C. CONTACT
STARTER COIL AND N.C. OVERLOAD CONTACT		ELECTROMAGNETIC SOLENOID		PILOT LIGHT		CONTACTS (TIME DELAY AFTER COIL)		HORN OR BUZZER	
						ENERGIZED		DE-ENERGIZED	
						N.O. CONTACT	N.C. CONTACT	N.O. CONTACT	N.C. CONTACT

Relay/Ladder Logic Symbols

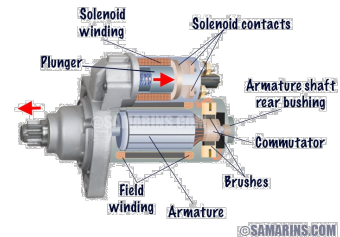
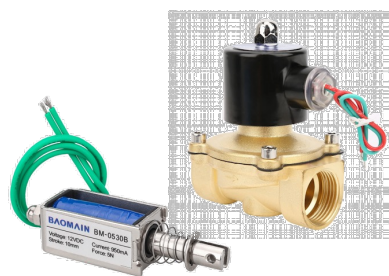
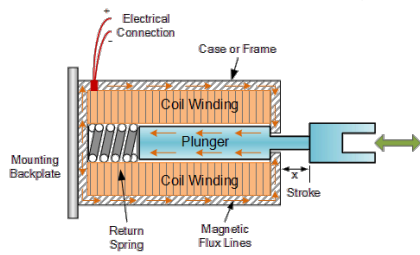
- Common Boolean logic gates can be represented using relay logic symbols
 - Normally-Open
 - Normally-Closed
 - Output Coil
 - This can be anything that produces a desired output, but for convenience we refer to a coil that is energized



<https://automationforum.co/plc-learning-series-4-how-a-ladder-logic-diagram-works-how-to-read-ladder-logic/ladder-logic-simples/>

Solenoid

- An electro-mechanical device that uses an energized coil to create a magnetic field, producing linear motion of a movable plunger
- It converts electrical energy into short-stroke mechanical force for actuation, switching, or control functions



<https://www.dr-solenoid.com/news/what-is-a-solenoid-and-how-does-it-work-and-application/>

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Summary

- Motion control systems may operate in open-loop (no feedback) or closed-loop (sensor feedback) configurations, with trade-offs in complexity, accuracy, and robustness
- Rotary shaft encoders (optical, magnetic, magneto-resistive) provide position information that enables precise closed-loop motor control
- Servo motors integrate motors, feedback, and control electronics to achieve accurate position control, commonly using PWM interfaces
- Brushed and brushless DC motors are combined with H-bridge circuits to control direction and speed via PWM
- Other electro-mechanical actuators—including linear actuators, solenoids, and mechanical or solid-state relays—extend embedded systems beyond rotary motion and enable controlled mechanical and electrical switching

Next Lecture

- RP2040 Stepper Examples
 - L298N H-Bridge
 - A4988 Driver

Questions

